



Glowing Horizon

A Fallout 4 Modlist & Survival Guide

V1.0.0 — Fallout 4 AE 1.11.221

A Fallout 4 Modlist & Survival Guide
Using Mod Organizer 2.5

Table of Contents

Installation Guide	4
About Glowing Horizon	4
Fallout 4 — Base Installation	4
Verifying Your Version	4
Mod Organizer 2.5	4
Download & Install	4
Initial Setup	5
Separators	5
Output Mods To Create	5
Register Tools in MO2	6
Installing F4SE Through MO2	6
MO2 Operating Rules	6
Installing Mods with MO2	7
Standard Mod Install	7
Load Order Basics	7
Left Pane Order	7
How to Mod Fallout 4	8
Install One Mod at a Time	8
Test Between Categories	8
Managing the Overwrite Folder	8
Patching Discipline	9
The Tools	9
Cyclic Workflow (per new mod)	9
Conflict Resolution in Practice	9
ModGroups	10
Cleaning Discipline	10
ESL-Flagged Patches	11
Risks	11
Recommended Folder Structure	11
Backing Up Your Save	12
Verifying Your Setup	12
Wave 0: Vault Dweller's First Light	13
The Narrative	13
Wave 0 Philosophy	13
How to Play: The Absolute Basics	13
Controls (PC, Keyboard & Mouse)	13
Controls (Gamepad, Xbox layout)	14
The SPECIAL System	14
Core Loop: Explore, Loot, Craft, Survive	14
Settlements, Briefly	14
Wave 0 Mods	16
Wave 1: Architect of the Commonwealth	24
The Narrative	24
Wave 1 Philosophy	24
Strategy: Settlements as a System	24
Deeper Crafting	24
Advanced Combat Notes	24

Wave 1 Mods	25
Appendix: Master Mod Reference	30
Wave 0 — Foundation	30
Wave 1 — Adds to Wave 0	30
Load Order Summary	30
Glossary	32

Installation Guide

About Glowing Horizon

Glowing Horizon is a curated modlist and comprehensive game guide for **Fallout 4**, version **1.11.221** (the Anniversary Edition / “AE” runtime, released May 27, 2026). It uses **Mod Organizer 2.5** to manage mods safely, without touching your base game files. The guide currently includes 23 verified mods across two waves.

The guide is split into two waves. Wave 0 takes complete beginners through a full how-to-play primer using nothing but vanilla systems plus interface, quality-of-life, and bugfix mods. Wave 1 builds on Wave 0 with new content and deeper settlement mechanics for players ready to go further.

Fallout 4 — Base Installation

1. Purchase Fallout 4 on Steam or GOG
2. Let Steam/GOG fully update the game — you want the **Anniversary Edition, version 1.11.221** (released May 27, 2026)
3. Install Fallout 4 outside Program Files (e.g. C:\Games\Fallout 4) to avoid Windows permission issues
4. Launch the game once through the vanilla launcher to generate your `Fallout4Prefs.ini` and `Fallout4.ini` files, then quit

Verifying Your Version

In the game’s main menu, check the bottom-left corner for the version string. It should read 1.11.221. If you see an older number (e.g. 1.10.163 “Old-Gen” or 1.10.984 “Next-Gen”), let Steam/GOG finish updating before installing any mods.

Warning: Fallout 4’s Anniversary Edition update (May 27, 2026) changed internal script offsets used by F4SE plugins. Every F4SE-dependent mod in this guide has been checked against the 1.11.221 runtime specifically — do not substitute older or unconfirmed F4SE plugin versions.

Mod Organizer 2.5

Mod Organizer 2 (MO2) is a virtual file system mod manager. It keeps every mod isolated in its own folder and merges them into a virtual view of your game’s Data folder at runtime — nothing is copied into your actual Fallout 4 installation. If something breaks, you uncheck the mod and the game sees its original files again.

MO2 operates across two panes. The **left pane** (mods list) controls which mod folders are active, and their priority order determines which copy of a conflicting file wins. The **right pane** (plugins tab) controls `.esp/.esm/.esl` load order — this is the order the game engine reads plugins in, and it operates independently of the left pane.

Download & Install

1. Get the latest MO2 2.5.x release from the official GitHub releases page or via Nexus Mods
2. Install MO2 as a standalone application to a location **outside** your Fallout 4 directory (e.g. C:\Modding\Tools\Mod Organizer 2) — this avoids Windows permission issues and keeps the toolchain separate from the game
3. Launch `ModOrganizer.exe`

Initial Setup

1. On first launch, create a **global instance**, select **Fallout 4**, and point it at your installation folder (e.g. C:\Games\Fallout 4)
2. Name the instance **Glowing Horizon** so it stays separate from any other Fallout 4 profile
3. If MO2 prompts for profile settings, tick every box — the profile should own its own INI files and game-specific settings rather than sharing global ones
4. Connect MO2 to Nexus Mods during first-launch setup and let MO2 handle nxm links — this makes MO2 your default download handler for “Mod Manager Download” buttons
5. Go to **Tools** → **Settings** → **General** and confirm the executable paths point to `Fallout4.exe` (or `f4se_loader.exe` once F4SE is installed)
6. Go to **Settings** → **Plugins** → **Diagnostics** and enable **Missing Data** and **Overwrite Contents** warnings — these catch mistakes early
7. In **Settings** → **Workarounds**, tick **Enable archives parsing (experimental)** so MO2 reports asset conflicts more reliably in the left pane

Separators

MO2 separators are labelled divider lines in the left pane that group your mods by category instead of one long flat list. They have no effect on the game — they are an organisational tool that makes a growing modlist navigable.

Create these separators in MO2 (right-click the left pane → **Create Separator**) before installing your first mod. The order here matches the guide’s recommended install sequence:

- **Baseline** — F4SE, Address Library, and engine-level frameworks
- **Bugfixes & Engine** — UFO4P, Buffout 4 NG, PRP, and engine-fix plugins
- **UI & Quality of Life** — MCM, FallUI series, Faster Loadscreens, XDI, Place Everywhere
- **Settlements & Content** — Workshop Framework, Sim Settlements 2, Homemaker, Scrap Everything
- **Crafting & Armor** — Armorsmith Extended, Faster Workshop AE
- **Weather & Visuals** — Vivid Weathers
- **Quests & New Content** — America Rising 2
- **Patches** — compatibility patches you or the community build
- **Output** — generated files from FO4Edit, BodySlide, and other tools

Tip: Separators cost nothing and save time when you have 20+ mods and need to find one quickly. Add them early.

Output Mods To Create

Before registering tools, create these empty mods under the Output separator so generated files never spill into the Overwrite folder:

- **FO4Edit Output** — manual patches and conflict-resolution plugins
- **F4SE Output** — F4SE log files and runtime-generated data
- **BodySlide Output** — generated meshes (if using BodySlide later)

Right-click a folder → **Create empty mod**, name it, and drag it under the Output separator. Leave it unchecked until something populates it.

Register Tools in MO2

Open MO2's executable editor (the gear icon in the top toolbar) and register the tools you will use throughout this guide. Use **Add from file...** to point MO2 at each tool's executable.

Tool	Executable	Arguments	Notes
F4SE	f4se_loader.exe	-forcesteamloader	Your default game launcher — use this for every launch after F4SE is installed.
L00T	L00T.exe	--game="Fallout4"	Sorts plugin load order. Run after adding new mods, before testing.
F04Edit	F04Edit.exe	-IKnowWhatImDoing -AllowMasterFilesEdit	Main conflict-review and manual-patching tool.
F04EditQAC	F04EditQuickAutoClean.exe	-IKnowWhatImDoing	Use only for plugins L00T explicitly flags for cleaning.

Tip: For F04Edit, also create a second entry called F04Edit – Conflicts with the arguments -veryquickshowconflicts -IKnowWhatImDoing. This is the entry you will use for the per-mod conflict-review workflow described in the Patching Discipline section below.

Installing F4SE Through MO2

F4SE (Fallout 4 Script Extender) is the scripting framework almost every mod in this guide depends on.

1. Download F4SE from Fallout 4 Script Extender (F4SE) — the page explicitly states support for game version **1.11.221**
2. Extract the archive. Copy f4se_loader.exe, f4se_1_11_221.dll, and f4se_steam_loader.dll (if present) directly into your Fallout 4 installation folder, **next to** Fallout4.exe — the loader must live alongside the game executable
3. In MO2, the rest of F4SE's files (the Data folder content) should be installed as a regular mod. Create an empty mod called F4SE Scripts under the Baseline separator and copy the archive's Data\Scripts folder into it
4. In MO2, go to the executables dropdown → **Edit Executables...**
5. Add a new executable named F4SE pointing at f4se_loader.exe in your Fallout 4 folder. Add the argument -forcesteamloader
6. Set F4SE as your default executable — always launch through F4SE, never through the vanilla Fallout4.exe or the Steam “Play” button

MO2 Operating Rules

- Launch the game through F4SE in MO2 every time, never through Steam or the stock launcher
- Run L00T after every major batch of new mods and before any testing session
- Treat the Overwrite folder (at the bottom of MO2's left pane) as an error state. Files only end up there when a tool wasn't configured to output into a dedicated mod. If you find files in Overwrite, move them into the appropriate output mod or create one

- Keep generated outputs in their own dedicated mods — never mix generated files into source mod folders
- Rebuild outputs after major changes to bodies, animations, or any other system that generates artifacts
- Create a clean backup profile in MO2 before any large installation batch so you can revert without reinstalling

Installing Mods with MO2

Standard Mod Install

1. Download a mod's archive (.7z / .zip / .rar) from Nexus Mods — use the **Mod Manager Download** button and let MO2 handle the nxm link
2. In MO2, the download appears in the **Downloads** tab (right pane). Right-click it → **Install**
3. If the mod uses a FOMOD installer, MO2 will show the installer wizard — choose the options specified in this guide's mod entries
4. The new mod appears in your left-hand **Mods** list, unchecked by default
5. Drag it into the appropriate separator group
6. Check the box next to the mod to activate it
7. For mods that add or change .esp/.esl plugins, switch to the **Plugins** tab (right panel) to set load order — see each mod's **Notes** for placement guidance

Load Order Basics

MO2's **Plugins** pane (right panel) controls plugin (.esp/.esm/.esl) load order, top to bottom. The game engine reads plugins in this order and later-loading records overwrite earlier ones — getting this order wrong is the single most common cause of mod conflicts.

General rules used throughout this guide:

- Masters and frameworks (Unofficial Fallout 4 Patch, Workshop Framework) load near the top, just after the base game masters and DLCs
- Bugfix and engine-fix mods load early
- Content and gameplay mods load after their required frameworks
- Weather mods load late to override exterior lighting and image-space records
- Patches and compatibility fixes always load **after** the mods they patch

Tip: MO2 2.5 includes a built-in LOOT integration (the sort icon in the **Plugins** tab toolbar). Running LOOT after installing several mods will auto-sort most of your load order correctly — but always double-check framework mods (Workshop Framework, Sim Settlements 2) and where their own documentation specifies.

Left Pane Order

The **left pane** (mods list) controls file-level priority. When two mods supply the same file (e.g. both replace the same texture or script), the mod lower in the left pane wins. This is separate from plugin load order — you can have a mod's files active even if its plugin loads early.

General left-pane rules:

- F4SE and engine frameworks at the top
- Bugfixes next, so their fixes override anything above
- UI mods after bugfixes
- Content mods next (meshes, textures, scripts)

- Patches at the very bottom so they override everything they patch

When MO2 shows a lightning bolt icon next to a mod, that mod is overriding (green bolt with +) or being overridden by (red bolt with -) another mod's files. Double-click the mod and check the **Conflicts** tab to see exactly which files conflict with which mods.

How to Mod Fallout 4

Modding is not just downloading and checking boxes. Every mod you add touches some part of the game — a record, a script, a texture, a mesh — and when two mods touch the same thing without a patch between them, one silently loses. The result might be invisible (a perk not applying) or catastrophic (a crash in a specific cell). The discipline described here prevents both.

Install One Mod at a Time

Install mods one at a time, never in bulk. After every new mod:

1. Install and activate the mod in MO2
2. Run LOOT to sort the updated plugin list
3. Launch the game through F4SE and spend 60 seconds testing — open the menu the mod touches, visit a settlement if it adds Workshop items, check that nothing obvious is broken
4. If the mod adds a plugin, run FO4Edit — Conflicts to check for new record conflicts (see Patching Discipline below)
5. Only after this passes do you move on to the next mod

This sounds slow. It is. But installing 15 mods at once and then trying to figure out which one broke the game takes far longer.

Test Between Categories

After completing an entire category (e.g. all Bugfixes & Engine mods, or all UI & Quality of Life mods), do a longer test:

1. Launch the game, load a clean save, and play for 15-20 minutes
2. Cover the systems the category touches — open every menu, build a Workshop item, fast-travel between cells, enter and exit interiors
3. If anything is wrong, the category boundary is a natural rollback point — disable the category's mods and re-enable them one by one to isolate the problem

Managing the Overwrite Folder

MO2's Overwrite folder (always visible at the bottom of the left pane) catches loose files that tools generate but have no designated output mod. Examples: F4SE logs, FO4Edit cache files, MCM settings exports.

- Empty the Overwrite folder after every tool run — either move its contents into the appropriate output mod or, if it's just logs, delete them
- If MCM settings files appear in Overwrite after configuring mods, create an MCM Settings mod under Output and drag them there
- If you see .esp files or .pexe cache files appearing in Overwrite, something is misconfigured — check your tool executable arguments

Warning: Leaving files in Overwrite means they load globally across every profile and mod combination. A stale F4SE log or a leftover MCM config can cause confusing behaviour that doesn't match any enabled mod. The Overwrite folder should be empty or contain only files you deliberately placed and understand.

Patching Discipline

Mod conflicts are not a sign something is broken. They are the natural result of two or more mods editing the same game record. The question is not whether conflicts exist — they always will — but whether the **right** mod wins each conflict.

Adopt the canonical mod-by-mod patching workflow: install one mod at a time, run FO4Edit in `-veryquickshowconflicts` mode, try load order first, build per-pair patches only when necessary, and use ModGroups to keep the conflict view clean.

The Tools

- FO4Edit (xEdit for Fallout 4) is the primary tool for inspecting, cleaning, and patching plugins. It loads your entire plugin list and visualises every record-level conflict.
- L00T is the load-order sorting tool. Run it after every new mod installation and before any FO4Edit session.

Download FO4Edit from FO4Edit on Nexus Mods and register it in MO2 as described in Register Tools above.

Cyclic Workflow (per new mod)

For every mod that adds or changes an `.esp/.esl` plugin, follow this cycle:

1. **Install and place.** Heuristic: broad world-changing mods load early; narrow local edits load late. Run L00T.
2. **Run FO4Edit with conflicts.** Launch your FO4Edit – Conflicts executable entry (registered with `-veryquickshowconflicts`). Right-click in the left-hand tree → **Hide no conflict and empty rows**. Now you only see records where at least two plugins compete.
3. **Try load order first.** Most conflicts are resolved by moving the winning mod later in the load order. Move the mod, re-run L00T, re-launch FO4Edit, and re-check. Only proceed to step 4 if moving the mod creates a different problem or the right mod genuinely cannot win through load order alone.
4. **Identify bad conflicts.** A bad conflict is one where the **loser** was supposed to win — for example, UFO4P fixes a quest record but a later-loading mod reverts it to the unfixed vanilla state. Focus only on these.
5. **Build a per-pair patch.** Right-click the losing record → **Copy as override into...** → `<new file>.esp` with the ESL template flag. Name it explicitly (e.g. `GlowingHorizon_UFO4P_x_Homemaker.esp`). Drag the corrected values from the losing plugin into the new patch record. Right-click the patch plugin → **Add Masters...** and add every mod whose records it references. Save into the FO4Edit Output mod.
6. **Build per-pair ModGroups.** After saving the patch, right-click each conflicting plugin pair → **Create ModGroup**. Move the generated `.ModGroup` files from MO2's `Overwrite` folder into each source mod's own MO2 folder. See the ModGroups section below for details.
7. **Re-open FO4Edit and confirm.** With the patch active and ModGroups applied, re-run the conflict view. The patch should resolve the conflicting records, and the ModGroups should hide the known-acceptable conflicts. You now have zero **visible** unresolved conflicts involving this mod.

Conflict Resolution in Practice

Not every conflict needs a patch. Here is how to decide:

- **Leveled lists (LVLI records):** Two mods adding items to the same leveled list. If both additions should appear in-game, they need a patch or a merged patch. If one is intentionally replacing the other, load order alone may suffice.

- **NPC records (NPC_ records):** Two mods editing the same NPC — one changes their inventory, the other changes their face. This almost always needs a patch, because whichever mod loads last will overwrite **all** fields, not just the ones it changes.
- **Quest records (QUEST records):** Complex and fragile. Never let two mods edit the same quest stage or alias without a patch. If you don't understand the conflict, err on the side of patching.
- **Cell records (CELL records):** Two mods placing objects in the same cell. A patch merges both sets of placements into a single cell record so both mods' objects appear. Without a patch, the winner's objects appear and the loser's are invisible.
- **Texture and mesh conflicts (left-pane only):** These do not appear in FO4Edit at all — they are file-level conflicts managed entirely by MO2's left-pane order. Check double-click → **Conflicts** tab for each mod in the left pane.

Tip: When in doubt about whether to patch, check the Nexus pages of both mods. Many popular mods publish compatibility patches — search for them before building your own. A community-tested patch is always safer than a hand-built one.

ModGroups

ModGroups are metadata files FO4Edit uses to hide conflicts you have **already reviewed and accepted**. They mark a specific pair of plugins as “known — the loser is supposed to lose,” so those conflicts no longer clutter the view.

Create a ModGroup after you have confirmed a conflict is intentional and acceptable:

1. In FO4Edit, right-click the losing plugin → **Create ModGroup**
2. FO4Edit generates a .ModGroup file and saves it to the Overwrite folder
3. Drag the .ModGroup file from Overwrite into the **losing** plugin's MO2 mod folder (not the winning one — the ModGroup tells FO4Edit “this plugin is expected to lose to that one”)

After building ModGroups for all accepted conflicts, re-opening FO4Edit should show you **only new, unresolved conflicts** — the view stays clean and actionable as your modlist grows.

Warning: Never create a ModGroup for a conflict you have not actually reviewed. A ModGroup that hides an unintended override is how a bug survives undetected through twenty more mod installs.

Cleaning Discipline

Some plugins ship with identical-to-master (ITM) records or deleted (UDR) references — remnants from the author's development process that can cause issues when other mods depend on the vanilla version of those records.

1. Run LOOT after installing a new mod
2. If LOOT flags a plugin as needing cleaning, launch your FO4EditQAC executable entry, select only that plugin, and wait for it to complete
3. After QAC finishes, select the plugin again in FO4Edit and run **Check for Errors** (right-click → **Check for Errors**). Red errors may indicate the mod is broken — reconsider using it
4. Never clean a plugin LOOT does not flag. Many mods ship ITMs intentionally as compatibility measures

The official Fallout 4 masters (Fallout4.esm, DLC .esm files) should also be cleaned. This is standard practice across the FO4 modding community:

1. Launch FO4EditQAC, select all vanilla masters **except** Fallout4.esm (clean it last, alone)
2. Repeat for each DLC master individually
3. Collect the cleaned .esm files into a Cleaned Vanilla Masters mod under the Baseline separator
4. Restore the originals from the FO4Edit Backups folder inside your Overwrite or FO4Edit directory

ESL-Flagged Patches

When building patches in FO4Edit, always use the ESL template flag (<new file>.esp with the ESL option checked). ESL-flagged plugins do not count toward Fallout 4's 254-plugin hard cap, but they still load and function exactly like regular plugins.

- Always use the [Template] ESL option when creating a new patch — never manually re-flag an existing ESP to ESL unless FO4Edit explicitly confirms it is safe
- Patches created with the ESL template are automatically compacted and flagged correctly
- ESL-flagged plugins share a limited address space (4096 slots across all light plugins) — this is rarely an issue for patch-sized plugins but avoid ESL-flagging mods that add hundreds of new records

Risks

- Keep your patches scoped — do not let one patch absorb every decision. A single monolithic patch is harder to update and harder to debug
- Date-stamp your patch filenames or maintain a changelog so you know when a source mod update requires rebuilding the patch
- Re-run FO4Edit after every source-mod update — an update to UFO4P or Workshop Framework can silently invalidate patches you built against an older version
- Patches that copy records from downstream plugins inherit unresolved conflicts — always check the full master chain, not just the two mods you are patching
- If you accumulate many small per-pair patches, consider merging them into themed patches (e.g. GlowingHorizon_Settlement_Patch.esp for all settlement-mod conflicts)
- Test that disabling one upstream source mod cleanly removes its dependent patch's masters — a patch that demands a master you have removed will crash the game on load

Tip: The discipline described here is the same one used by every major Skyrim and Fallout 4 modlist. It is an investment of time at the start that pays back every time you install a new mod without breaking anything, and every time you debug a problem in minutes instead of hours because the conflict view is clean.

Recommended Folder Structure

```
C:\Games\Fallout 4\          <- vanilla game install (untouched)
├─ Fallout4.exe
├─ f4se_loader.exe           <- F4SE, installed directly here
├─ f4se_1_11_221.dll
└─ Data\                     <- only vanilla + DLC files live here

C:\Modding\Tools\Mod Organizer 2\ <- MO2 instance (separate location)
├─ ModOrganizer.exe
├─ mods\                     <- every mod's isolated folder
│   └─ Baseline\
│       └─ F4SE Scripts\
```

```

|   |   | Bugfixes & Engine\
|   |   | | Unofficial Fallout 4 Patch\
|   |   | | ...
|   |   | | UI & Quality of Life\
|   |   | | Settlements & Content\
|   |   | | Output\
|   |   | | | F04Edit Output\
|   |   | | | F4SE Output\
|   |   | |
|   |   | profiles\
|   |   | | Glowing Horizon\
|   |   | | | plugins.txt      <- load order
|   |   | | | modlist.txt     <- mod activation state
|   |   | |
|   |   | downloads\         <- archived mod downloads
|   |   | overwrite\         <- loose generated files

```

Backing Up Your Save

Before installing any mods, make a manual backup of your Documents\My Games\Fallout4\Saves folder. MO2 does not touch your saves, but it is good practice before any modding project.

Also back up your MO2 profile periodically: copy the entire profiles\Glowing Horizon\ folder to a safe location. This preserves both your plugins.txt (load order) and modlist.txt (mod activation state) — recovering from a broken load order is trivial if you have this backup.

Verifying Your Setup

Before beginning the Wave 0 modlist:

1. Launch the game through MO2's **F4SE** executable entry
2. Watch for the F4SE version splash in the main menu's bottom-right corner (in addition to the game version 1.11.221 in the bottom-left)
3. Start a new game and verify the vanilla opening sequence runs without crashing — this confirms F4SE and the base game are in agreement before any mods are added
4. Quit, check MO2's Overwrite folder is empty or contains only expected log files
5. You are now ready to begin installing Wave 0 mods

Wave 0: Vault Dweller's First Light

“War... war never changes. But you're about to.”

The Narrative

One hundred and eleven years. That's how long the cryo-pod in Vault 111 has held you frozen, dreaming static dreams while the world outside burned, rebuilt, and burned again. When the pod hisses open and the lid swings up, the vault's stale air hits you like a slap. Your neighbors' pods are open and empty. Something is very wrong, and there is nobody left to ask.

You climb out with nothing but the clothes on your back, a Pip-Boy strapped to a stranger's wrist that is now yours, and a single, burning question: where is your family? The vault's exit door grinds open on a Commonwealth you don't recognize — green light, cracked highways, and the distant howl of something that used to be a dog.

You are not a soldier. You are not a scavenger. You are a person who woke up too late, in a world too broken, with everything still to learn.

Wave 0 Philosophy

Wave 0 is a **vanilla-plus** experience — the game exactly as Bethesda built it, sharpened with interface, bugfix, and quality-of-life mods only. Nothing here changes combat balance, adds new areas, or introduces new mechanics.

- **What changes:** menus, HUD clarity, bug fixes, script-engine stability
- **What stays untouched:** enemy stats, loot tables, quest content, world design

This wave exists to teach you Fallout 4 as designed, without the friction of its rougher engine quirks.

How to Play: The Absolute Basics

Controls (PC, Keyboard & Mouse)

Action	Default Key
Move	W A S D
Sprint	Shift (hold)
Jump	Space
Crouch / Sneak	Ctrl
Fire weapon	Left Mouse Button
Aim (VATS-free aiming)	Right Mouse Button
V.A.T.S.	Q
Reload	R
Switch weapon	1-8 or Mouse Wheel
Pip-Boy	Tab
Map (inside Pip-Boy)	Tab then select Data tab
Activate / Talk / Loot	E
Holster weapon	R (hold) or via Pip-Boy
Power Armor exit	E near open frame
Quick save / load	F5 / F9

Controls (Gamepad, Xbox layout)

Action	Xbox Button
Move	Left stick
Look / Aim	Right stick
Sprint	Left stick click
Jump	A
Crouch / Sneak	Right stick click
Fire weapon	RT
V.A.T.S.	LT
Reload	X
Switch weapon	Y (favorites wheel)
Pip-Boy	Menu / View
Activate / Talk / Loot	A

The SPECIAL System

Every character is built from seven core attributes — **S.P.E.C.I.A.L.**:

Attribute	Governs
Strength	Melee damage, carry weight, some Power Armor perks
Perception	V.A.T.S. accuracy, detecting traps/enemies, lockpicking perks
Endurance	Max HP, action point regen, chem resistance perks
Charisma	Dialogue/persuasion checks, companion affinity, settlement happiness perks
Intelligence	XP gain rate, hacking, science/crafting perks
Agility	Action Points, sneak effectiveness, reload/attack speed perks
Luck	Critical hit chance/recharge, random loot quality, gambling perks

Each attribute unlocks a column of perks in the Pip-Boy's Perks chart — the higher the attribute, the deeper into that column you can invest as you level up. There's no wrong build for a first playthrough: put points where your intended playstyle wants them (Agility/Perception for a sneaky rifle build, Strength/Endurance for melee and heavy armor, and so on).

Core Loop: Explore, Loot, Craft, Survive

- **Explore** the Commonwealth's landmarks, following quest markers or just wandering — almost everything is hand-placed and worth investigating
- **Loot** everything you can carry (within your Strength-based carry weight); junk items are broken down into components at workbenches
- **Craft** at Workbenches: Weapons Workbench, Armor Workbench, Chemistry Station, Cooking Station, and the Power Armor Station
- **Survive** by managing hunger/thirst/sleep only if you enable Survival difficulty later — on Normal/Hard, health and rest are less punishing but still matter

Settlements, Briefly

Early in the main quest you'll gain access to the Workshop system at Sanctuary Hills. Press the Workshop-mode key (default Tab-adjacent, check your control scheme) while standing in a

settlement to build. Wave 0 keeps settlement building vanilla — deeper settlement tools arrive in Wave 1.

Wave 0 Mods

These mods form the essential, stability-first foundation for the rest of this modlist. Every mod below has been verified against the 1.11.221 (Anniversary Edition) runtime.

Framework

Fallout 4 Script Extender (F4SE)

Wave: 0 (baseline, required by nearly everything below)

Dependencies: None

MO2 Install: Extract directly into your Fallout 4 game folder (next to Fallout4.exe), NOT into MO2's mods folder. Add f4se_loader.exe as an MO2 executable and always launch through it.

What it does: A modder's resource that expands Fallout 4's Papyrus scripting engine, exposing new functions without altering the game's executable on disk.

Why: Nearly every mod in this guide beyond simple ESP edits requires F4SE as a hard dependency.

Notes: The mod page explicitly states game version 1.11.221 is required for the current F4SE build. Always match your F4SE version to your exact game version after any Bethesda patch.

Framework

Address Library for F4SE Plugins

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Install as a regular mod via MO2 (drag-and-drop archive). No plugin/ESP — it's a pure F4SE resource.

What it does: Provides a version-independent lookup table of memory addresses that other F4SE plugins use internally, so plugin authors don't hard-code offsets that break on every game patch.

Why: A hard requirement for most other F4SE plugins in this list (Buffout 4 NG, Faster Workshop AE, etc.).

Notes: Tags confirm "Works with Next-Gen Update" and "Anniversary Update Compatible"; the main file is explicitly versioned 1.11.221.

Bugfix

Unofficial Fallout 4 Patch (UFO4P)

Wave: 0

Dependencies: None (DLC-aware — install the DLC you own)

MO2 Install: Install via MO2's FOMOD installer; select the DLC modules you actually own. Load near the top of your plugin list, right after the base game masters.

What it does: A community-maintained compilation fixing thousands of bugs Bethesda never officially resolved — from quest-breaking issues to minor visual glitches.

Why: The foundational bugfix mod almost every other Fallout 4 mod list builds on top of.

Notes: Version 2.2.1a (28 May 2026). Tags confirm "Works with Next-Gen Update" and "Anniversary Update Compatible". Load order: near the very top, just under Fallout4.esm and DLC masters.

Framework

Mod Configuration Menu (MCM)

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Install via MO2 as a regular mod. Adds a new "Mod Configuration" entry to the in-game pause menu.

What it does: Provides a centralized, in-game settings screen that other mods hook into instead of shipping their own ad-hoc holotape or INI-based configuration.

Why: Required by many Wave 1 mods (Sim Settlements 2, Faster Workshop AE) for their settings screens.

Notes: Tags confirm "Works with Next-Gen Update" and "Anniversary Update Compatible"; the current main file is explicitly listed "For Fallout 4 game version 1.11.221".

Bugfix

Buffout 4 NG with PDB support

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE), Address Library for F4SE Plugins

MO2 Install: Install via MO2; comes with a Toml config file (Buffout4.toml) — leave defaults unless you know what you're changing.

What it does: An engine-level stability plugin providing crash logging and a wide set of low-level bug fixes for the game's engine, not just Papyrus scripts.

Why: Dramatically reduces random crashes-to-desktop and is the current standard replacement for the original (Next-Gen-broken) Buffout 4.

Notes: Version 1.38.2, updated June 2026 (a month after the 1.11.221 release), and its description states the single-DLL NG build "should work in flat (pre and post Next Gen) and VR." No explicit AE compatibility tag was found on the page itself during this research pass — treat as best-available evidence rather than a guaranteed tag match, and watch its comments for AE-specific issue reports.

UI/QoL

Faster Loadscreens

Wave: 0

Dependencies: None

MO2 Install: Simple archive install via MO2, no FOMOD choices needed.

What it does: Speeds up loading screens up to 3x and swaps the animated 3D loadscreen models for lightweight 2D backgrounds, while keeping the original loading tip text and level display.

Why: One of the highest quality-of-life wins for a beginner who will be fast-traveling and loading saves constantly while learning the game.

Notes: Description explicitly states "Supports VR/OG/NG/AE" — safe across every current game version including 1.11.221.

Bugfix

Empty Vendor List Bug Fix

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Simple archive install via MO2.

What it does: Fixes a long-standing engine bug where vendor barter menus occasionally appear completely empty until you back out and re-open them.

Why: A small but persistent annoyance in vanilla trading, especially noticeable while learning the game's economy.

Notes: Explicit "Supports Game versions: OG (1.10.163), NG (1.10.984), AE (1.11.137-1.11.221)" on the file page.

Bugfix

Magic Effect and Spell Engine Fixes

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Simple archive install via MO2.

What it does: An F4SE plugin fixing multiple engine-level bugs in how magic effects (chems, radiation, status effects) are applied and removed.

Why: Improves the reliability of chem, radiation, and status-effect behavior that the vanilla engine mishandles in edge cases.

Notes: Explicit "Supports Game versions: OG (1.10.163), NG (1.10.984), AE (1.11.137-1.11.221)" on the file page.

Bugfix

Motion Vector Fixes

Wave: 0

Dependencies: None

MO2 Install: Simple archive install via MO2.

What it does: Corrects incorrect motion vector data output by the engine, improving the accuracy of motion blur, upscalers (DLSS/FSR), and frame-generation technology.

Why: A meaningful visual-quality and performance fix for anyone using modern upscaling or frame generation, at zero gameplay risk.

Notes: Description states "Supports all game versions and ENB."

Bugfix

Companion Shoots At Player Fix

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Simple archive install via MO2.

What it does: Fixes an engine bug where companion NPCs occasionally hit the player character with friendly fire during chaotic firefights.

Why: Removes a frustrating source of unfair damage for players still learning combat positioning around companions.

Notes: Explicit "Supports Game versions: OG (1.10.163), NG (1.10.984), AE (1.11.137-1.11.221)" on the file page.

Bugfix

Previsibines Repair Pack (PRP) Stable Branch

Wave: 0

Dependencies: Unofficial Fallout 4 Patch (hard requirement — you WILL crash in specific cells without it)

MO2 Install: Install via MO2 (multi-part archive — install both parts as directed on the mod page). Load directly after Unofficial Fallout 4 Patch, near the very top of your plugin list.

What it does: A full rebuild of the game's precombined meshes and previsibility/occlusion data, fixing vanilla rendering bugs and reducing draw calls across the Commonwealth for a real performance gain.

Why: One of the most load-bearing stability and performance mods in the entire Fallout 4 modding ecosystem; many later mods (including Scrap Everything and Sim Settlements 2 content) assume it's present and provide compatible patches built against it.

Notes: Build 81.3, actively patched by the community through 2026 (e.g. "A Collection of Kuz's PRP Patches", built against PRP 81, itself tagged Anniversary Update Compatible). Load order is critical — must load early, right after UFO4P.

Extended Dialogue Interface (XDI)

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Simple archive install via MO2. No ESP — pure F4SE engine patch.

What it does: Removes the vanilla engine's hardcoded 4-option dialogue limit and adds proper engine support for any number of player dialogue choices, without altering any vanilla dialogue.

Why: A safe, content-neutral engine fix; many Wave 1 quest/content mods (including Sim Settlements 2) rely on it for their dialogue trees to display correctly.

Notes: The mod's file history includes a build explicitly labeled "For Fallout 4 AE update 3, game version 1.11.221."

FallUI - HUD

Wave: 0

Dependencies: Mod Configuration Menu (MCM)

MO2 Install: Install via MO2 as a regular mod. Load after all other HUD mods. The in-game layout manager lives under MCM's Mod Configuration menu — select a pre-made layout or enter Edit Mode to customise widget positions, colours, and sizes.

What it does: A highly configurable HUD framework. Every on-screen widget — compass, health and action-point bars, ammo counter, XP bar, quest tracker, hit markers, stealth indicator — can be repositioned, resized, and styled from an in-game layout editor. Layouts can be shared, imported, and exported.

Why: The vanilla HUD wastes screen real estate and buries critical information behind cramped console-first design. FallUI - HUD gives Wave 0 players a cleaner, PC-optimised interface without touching a single gameplay variable.

Notes: The mod page carries the explicit "Works with Next-Gen Update" tag. For item icons in the HUD, pair with FallUI - Inventory and an item sorter (FIS, VIS, VIS-G). Archive invalidation must be enabled.

FallUI - Inventory

Wave: 0

Dependencies: FallUI - HUD (recommended, for icon tags on the HUD)

MO2 Install: Install via MO2's FOMOD installer, selecting your preferred item-sorter integration. Mid-game-safe — no changes made to savegames. Load order: item tag/icon mod → FallUI - Inventory → FallUI - HUD.

What it does: A full overhaul of the Pip-Boy, container, barter, workshop, and companion inventory interfaces. Adds multi-column sortable lists, side-by-side item comparison, bulk container transfer, automatic category grouping, and configurable display density — everything the vanilla console-first inventory was never designed to do.

Why: The vanilla inventory is the single highest-friction interface in Fallout 4 on PC. FallUI - Inventory transforms it into a proper desktop-UI experience without altering any item stats, balance, or loot tables.

Notes: The mod page carries the explicit "Works with Next-Gen Update" tag. For sorted items with icons, pair with an item sorter (FIS, VIS, VIS-G). Works out-of-the-box if you already use DEF_INV or DEF_UI. Increase Pip-Boy resolution (1920×1080 or higher via INI tweak) for best results with multi-column layouts.

UI/QoL

Place Everywhere

Wave: 0

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Simple archive install via MO2. Pure F4SE plugin — no ESP needed, injects via the script extender. Toggle features via the in-game F8 menu.

What it does: An F4SE plugin that removes placement restrictions in workshop mode, letting you place objects inside other objects, in water, in mid-air, and anywhere else the vanilla engine normally blocks with a red outline. Also makes most scrap-only (yellow-outline) objects selectable and movable.

Why: The vanilla workshop placement rules are overly restrictive — they block creative building without improving balance. Place Everywhere gives Wave 0 players real control over settlement construction without adding any new objects, quests, or mechanics.

Notes: Version 1.22.1.1221 (3 June 2026), explicitly built for game version 1.11.221. Tags confirm both "Works with Next-Gen Update" and "Anniversary Update Compatible." Post-Main-Release Hotkeys and Extended-Object-Selection are disabled by default — enable via the F8 menu if desired.

Wave 0 Completion Checklist

- You can navigate the Pip-Boy's Stat, Inv, Data, and Radio tabs confidently
- You understand your SPECIAL build and have spent at least 3 perk points
- You've built a basic settlement structure at Sanctuary Hills
- You've completed at least one full dungeon (e.g. Corvega Assembly Plant or Concord's Museum of Freedom)
- Your load order matches the order above, and the game has launched via F4SE without crashing

Wave 1: Architect of the Commonwealth

“A settlement isn’t just walls and a bed. It’s a promise that tomorrow is worth building for.”

The Narrative

Months have passed since the vault door opened on a ruined world. You’ve learned to read the Commonwealth’s dangers, to loot smart, to survive a firefight without panicking. Sanctuary Hills is no longer just a ruin with your name on the deed — it’s the first real foothold you’ve built, and word is spreading.

Settlers are arriving on their own now, drawn by rumors of a place with working turrets and a roof that doesn’t leak. The Minutemen general’s coat you inherited feels a little less like someone else’s uniform. You start noticing things you didn’t before: a synth watching too closely from a doorway, a settlement’s crops failing quietly, a trade caravan that never quite makes it through.

It’s time to build something bigger than survival. It’s time to build a home worth defending — and to start shaping the Commonwealth instead of just passing through it.

Wave 1 Philosophy

Wave 1 keeps every Wave 0 mod active and adds **new content and new mechanics** — deeper settlement building, expanded crafting, and quality-of-life tools for managing a growing empire of settlements. Nothing here is a pure power spike: these mods add systems and options, not bigger numbers.

- **What’s new:** settlement building depth, crafting/armor systems, workshop performance tools
- **What’s still off the table:** god-mode items, all-knowing detection mods, cheat menus

Strategy: Settlements as a System

By this point you should think of settlements less as decoration and more as a network:

- **Happiness** drives settler productivity and unlocks certain worker assignments — keep food, water, beds, and defense above the settlement’s population count
- **Supply lines** (assign a settler as a Provisioner between two settlements) share workshop resources across your network, letting you build with materials stockpiled anywhere
- **Defense rating** should meaningfully exceed your threat level in the area, or you’ll face costly raids
- **Local Leader** (Charisma perk) unlocks supply lines and a wider crafting recipe pool at the Chemistry/Cooking stations — a near-mandatory perk investment for this wave

Deeper Crafting

With an Armor Workbench and the mods below, armor pieces can be broken down into more granular components, letting you mix-and-match cosmetic slots (e.g. wearing a hat under a helmet, or a coat under power armor) far more flexibly than the vanilla keyword system allows.

Advanced Combat Notes

- Legendary enemies (marked with a star icon on their health bar) always drop a Legendary item — worth actively hunting once you’re geared for the fight
- V.A.T.S. critical hits are banked via Luck-based critical charge; time your V.A.T.S. usage around a full charge meter for burst damage
- Power Armor consumes fusion cores continuously while worn and moving — always carry spares before a long expedition

Wave 1 Mods

These mods build on every Wave 0 mod above. All have been verified against the 1.11.221 (Anniversary Edition) runtime.

Framework

Workshop Framework

Wave: 1

Dependencies: Fallout 4 Script Extender (F4SE)

MO2 Install: Install via MO2's FOMOD installer. Load directly after Unofficial Fallout 4 Patch, before any mod that requires it (Sim Settlements 2, Faster Workshop AE).

What it does: Opens up the settlement/workshop system for the modding community — faster, more stable workshop scripts, and new resource types other mods can build on.

Why: The load-bearing framework for the settlement-focused mods in this wave; required by Sim Settlements 2.

Notes: Version 2.5.0, updated July 2026. Tag confirms "Works with Next-Gen Update." No pure content changes on its own — it's infrastructure.

Content

Sim Settlements 2

Wave: 1

Dependencies: Workshop Framework, Fallout 4 Script Extender (F4SE)

MO2 Install: Install the main file via MO2's FOMOD installer. Load after Workshop Framework and Unofficial Fallout 4 Patch. Start the associated quest from a radio signal once in-game to begin the tutorial.

What it does: Adds an entirely new settlement-building layer: settlers construct and upgrade their own plots automatically based on templates you (or the community) design, plus an original questline explaining why settlements matter to the wider Commonwealth.

Why: Turns settlement building from a manual chore into an emergent system, and gives Wave 1's "architect" narrative real mechanical teeth without granting the player character any new combat power.

Notes: Version 3.6.1, updated July 2026. Tag confirms "Anniversary Update Compatible" (visible on the Chapter 3 sub-page for the same mod family). Very large download — the All Chapters Pack is several GB.

Mechanics

Scrap Everything

Wave: 1
Dependencies: None

MO2 Install: Simple archive install via MO2. Load after Unofficial Fallout 4 Patch and any precombine/previs-related mods.

What it does: Allows scrapping of static clutter, foliage, and structural pieces in settlements that are normally locked in place — grass, garbage piles, fences, entire pre-existing buildings.

Why: Gives real control over how a settlement looks and is laid out, which matters once you're building at the scale Sim Settlements 2 encourages.

Notes: Tag confirms "Anniversary Update Compatible." Can affect precombined meshes in some cells; since Wave 0 already installs Previsibines Repair Pack (PRP), check for a location-specific PRP compatibility patch before reporting any visual glitch as broken.

Mechanics

Faster Workshop AE (F4SE)

Wave: 1
Dependencies: Fallout 4 Script Extender (F4SE), Address Library for F4SE Plugins

MO2 Install: Simple archive install via MO2.

What it does: An F4SE plugin that removes the artificial delay when entering/exiting Workshop build mode and speeds up the loading of build-menu categories.

Why: Sim Settlements 2 and Scrap Everything both make you spend far more time in Workshop mode than vanilla — this mod removes the friction of that increased time investment.

Notes: Explicit "Supports the Anniversary Editions (1.11.137.0 and later)" on the file page — directly covers 1.11.221.

Content

Armorsmith Extended

Wave: 1

Dependencies: None (standalone as of v7.0; older versions required AWKCR — this list uses the standalone v7.0)

MO2 Install: Simple archive install via MO2's FOMOD installer. Load late in your plugin order, after any other armor/clothing content mods it might need to patch.

What it does: Expands armor and clothing slot compatibility, letting you layer outfits, hats, glasses, and armor pieces in combinations the vanilla keyword system normally blocks — plus new crafting recipes for existing armor.

Why: Directly supports Wave 1's deeper-crafting focus: it does not add new stats or power, only more flexible ways to wear and craft what already exists.

Notes: Version 7.0, updated January 2026. Tags confirm "Works with Next-Gen Update" and "Anniversary Update Compatible."

Graphics

Vivid Weathers - Fallout 4 Edition

Wave: 1

Dependencies: None

MO2 Install: Install via MO2's FOMOD installer. Select your preferred colour palette and season preset during installation. Includes an in-game holotape (craftable at the Chemistry Station) for adjusting weather chances and colour temperature on the fly.

What it does: A comprehensive weather and climate overhaul adding dozens of new weather types — volumetric fog, enhanced cloud systems, seasonal colour grading, and dynamic night-time darkness. Replaces the entire vanilla weather pool with a richer, more atmospheric palette of storm, fog, clear, and overcast conditions.

Why: The Commonwealth's visual identity is its strongest asset, yet the vanilla weather system cycles through a small pool of near-identical grey-skies conditions. Vivid Weathers replaces the game's visual backdrop without touching gameplay balance, loot tables, or combat.

Notes: The mod page carries the explicit "Works with Next-Gen Update" tag. If using an ENB preset, select the ENB-compatible installation option in the FOMOD and disable the mod's built-in depth-of-field. Load late in your plugin order — weather mods should win conflicts against any other mod touching exterior lighting or image-space records.

Content

America Rising 2 - Legacy of the Enclave

Wave: 1

Dependencies: Extended Dialogue Interface (XDI)

MO2 Install: Install via MO2's FOMOD installer. Load after Sim Settlements 2 and any other large quest or content mods.

What it does: A fully voiced, multi-mission questline that lets the player join — or oppose — a resurgent Enclave faction operating from a new worldspace. Adds new Enclave-themed NPCs, weapons, armour, a player-operable base, and branching story decisions that integrate with the vanilla main quest and faction system.

Why: The Enclave is Fallout's most iconic antagonist faction, and the vanilla game gives them almost no presence in the Commonwealth. America Rising 2 fills that gap with a questline that meshes mechanically with Wave 1's settlement-network narrative — the Enclave's industrial projects require a functioning supply chain of player settlements.

Notes: The mod page carries the explicit "Works with Next-Gen Update" tag and is actively patched through 2026. This mod adds to the vanilla main quest rather than replacing it — no alternative-start or faction-replacement mods are required.

Content

Homemaker - Expanded Settlements

Wave: 1

Dependencies: All official DLC (Automatron, Wasteland Workshop, Far Harbor, Contraptions Workshop, Vault-Tec Workshop, Nuka-World)

MO2 Install: Install via MO2's FOMOD installer, selecting the DLC modules you own. Load after Scrap Everything in your plugin order.

What it does: Adds over a thousand new buildable objects to the settlement Workshop menu — furniture sets, structural pieces, plantable foliage, decorative clutter, and modular building components inspired by pre-War, military, industrial, and wasteland architectural styles. Includes clean, restored, and weathered variants of most pieces.

Why: Sim Settlements 2 handles the macro-scale settlement automation — settlers building and upgrading their own plots. Homemaker handles the micro-scale, giving Wave 1's architect the full palette of buildable assets needed to make every settlement feel distinct and hand-crafted rather than a grid of identical SS2 plots.

Notes: The mod page carries both "Works with Next-Gen Update" and "Anniversary Update Compatible" tags. Version 1.79.6 (June 2025). Menu injection can add a few seconds of load time to the Workshop

menu on first open per session — lighter than most equivalent content packs and well within tolerance with Faster Workshop AE active.

Wave 1 Completion Checklist

- You have Local Leader rank 2 and at least one active supply line between settlements
- You've built or adopted a Sim Settlements 2 plot in at least two settlements
- You've completed the Sim Settlements 2 introductory questline
- Your settlement defense rating meaningfully exceeds the local threat level
- You've crafted at least one layered outfit using Armorsmith Extended's expanded slots

Appendix: Master Mod Reference

A single reference table of every mod in this guide, in recommended load-order-adjacent grouping. Mods accumulate across waves — Wave 1 includes everything from Wave 0.

Wave 0 — Foundation

Mod	Category	Nexus ID
Fallout 4 Script Extender (F4SE)	Framework	42147
Address Library for F4SE Plugins	Framework	47327
Unofficial Fallout 4 Patch (UFO4P)	Bugfix	4598
Previsibines Repair Pack (PRP) Stable Branch	Bugfix	46403
Mod Configuration Menu (MCM)	Framework	21497
Buffout 4 NG with PDB support	Bugfix	64880
Faster Loadscreens	UI/QoL	102233
Empty Vendor List Bug Fix	Bugfix	83252
Magic Effect and Spell Engine Fixes	Bugfix	83433
Motion Vector Fixes	Bugfix	98544
Companion Shoots At Player Fix	Bugfix	81758
Extended Dialogue Interface (XDI)	UI/QoL	27216
FallUI - HUD	UI/QoL	51813
FallUI - Inventory	UI/QoL	48758
Place Everywhere	UI/QoL	9424

Wave 1 — Adds to Wave 0

Mod	Category	Nexus ID
Workshop Framework	Framework	35004
Sim Settlements 2	Content	47976
Scrap Everything	Mechanics	5320
Faster Workshop AE (F4SE)	Mechanics	100532
Armorsmith Extended	Content	2228
Vivid Weathers - Fallout 4 Edition	Graphics	15466
America Rising 2 - Legacy of the Enclave	Content	75767
Homemaker - Expanded Settlements	Content	1478

Load Order Summary

1. Fallout4.esm + official DLC masters
2. Unofficial Fallout 4 Patch (UFO4P)
3. Previsibines Repair Pack (PRP) Stable Branch
4. Workshop Framework
5. Sim Settlements 2 (+ chapters, if using more than the base chapter)
6. America Rising 2 - Legacy of the Enclave
7. Armorsmith Extended
8. Homemaker - Expanded Settlements
9. Scrap Everything

10. Vivid Weathers - Fallout 4 Edition (weather mods load late to override exterior lighting and image-space records from other mods)
11. Everything else (pure F4SE plugins with no ESP: F4SE itself, Address Library, Buffout 4 NG, Faster Loadscreens, Empty Vendor List Bug Fix, Magic Effect and Spell Engine Fixes, Motion Vector Fixes, Companion Shoots At Player Fix, Faster Workshop AE, XDI, MCM, Place Everywhere, FallUI - HUD, FallUI - Inventory — these either inject via F4SE or are pure UI/SWF overrides with no load-order position relative to plugins)

Note: Always run MO2's LOOT integration after adding new mods, and manually verify framework mods (Workshop Framework, Sim Settlements 2) sit where their own documentation specifies before trusting an automated sort blindly.

Glossary

Term	Definition
AE	Anniversary Edition — the Fallout 4 runtime branch starting at game version 1.11.137, culminating in 1.11.221 (May 27, 2026), this guide’s target version.
AWKCR	Armor and Weapon Keywords Community Resource — a legacy shared-keyword framework many older armor mods required; largely superseded by standalone mods like Armorsmith Extended v7.0+.
ESP / ESM / ESL	Plugin file types Fallout 4 loads: ESM (master, foundational), ESP (plugin, standard mod), ESL (light plugin, uses a smaller ID range so more can be loaded at once).
F4SE	Fallout 4 Script Extender — a modder’s resource that extends the Papyrus scripting engine’s capabilities without modifying the game executable.
FOMOD	A packaged mod installer format with a step-by-step options wizard, supported natively by Mod Organizer 2.
Legendary Enemy	An enemy marked with a star icon on its health bar; always drops a Legendary-quality weapon or armor piece on death.
LOOT	Load Order Optimisation Tool — sorts plugin load order automatically based on community-maintained masterlist rules; built into MO2 2.5’s Plugins tab.
Load Order	The sequence in which plugins (ESM/ESP/ESL files) are read by the game engine; later-loaded plugins override earlier ones for the same records.
MCM	Mod Configuration Menu — an in-game settings screen (accessed from the pause menu) that mods hook into instead of shipping ad-hoc configuration tools.
MO2	Mod Organizer 2 — a virtual file system mod manager that keeps every mod isolated until merged at runtime, leaving the base game files untouched.
NG	Next-Gen — the Fallout 4 runtime branch starting at game version 1.10.984 (2024), preceding the Anniversary Edition (AE).
OG	Old-Gen — the original, pre-2024 Fallout 4 runtime, version 1.10.163 and earlier.
Papyrus	Fallout 4’s built-in scripting language, used for quests, mod logic, and gameplay systems.
Perk	A gameplay bonus unlocked with level-up points, gated behind a SPECIAL attribute’s rank (e.g. the Gunslinger perk requires a minimum Agility score).
Precombine / Previs	Engine-level optimization data (precombined meshes and previsibility/occlusion culling) that lets the game render busy areas efficiently; scrapping/building mods can break this data unless a compatible patch (e.g. a Previsibines Repair Pack build) exists.

Term	Definition
Provisioner	A settler assigned to walk a supply line between two settlements, sharing workshop resources between them; requires the Local Leader perk to assign.
SPECIAL	The seven core character attributes: Strength, Perception, Endurance, Charisma, Intelligence, Agility, Luck.
V.A.T.S.	Vault-Tec Assisted Targeting System — a tactical time-slow targeting mode that spends Action Points to queue precise, chance-based attacks.
Workshop / Settlement Mode	The build interface used to place structures, furniture, and defenses within a settlement's workshop boundary.

